

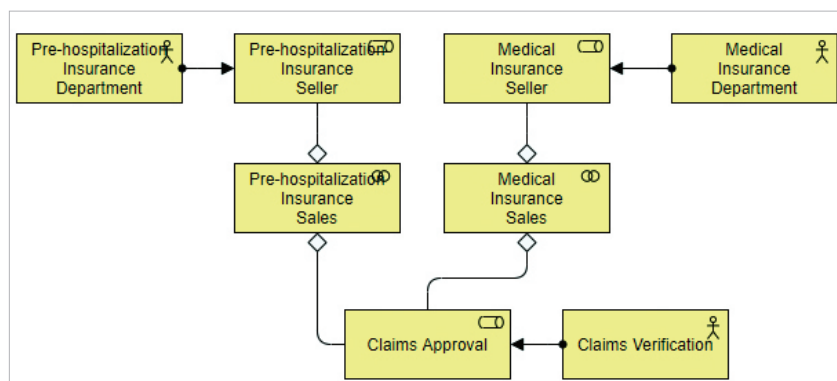


Figure 3: Blockchain based processing of insurance claims

With a blockchain based solution, we can implement smart contract-based solutions that send alerts in case of fluctuations in these readings so that hospitals can remotely take active measures to handle any emergency.

Processing of insurance claims:

Real-time processing of medical insurance for patients is a tedious task as it involves verification of documents and invoices/bills, hospital reports, as well as insurance coverage and policy rules. Using the blockchain platform, we can enable multiple approvers across the nodes for digital document processing so that insurance claims can be settled with ease.

Electronic medical records:

Sharing of electronic medical records (EMRs) and electronic health reports (EHRs) between different hospitals is a very important need being fulfilled by the blockchain platform. When patients are hospitalised, getting a report of their past ailments is important to proceed with diagnosis and further treatment.

This is a time-consuming task, as it needs collaboration between hospitals to transfer the required data and involves internal approvals as well. This can be achieved with a blockchain and IoT based solution, that can help search a patient's medical records and treatment history, prescriptions given for previous ailments, etc. Action can then be taken to ensure that the

entire EMR transmission is automated completely without compromising security while transmitting records to unauthorised parties by using blockchain based security services.

Challenges in the medical supply chain

Supply chain management is one of the biggest costs of healthcare providers. According to a Cardinal Health and SERMO Intelligence survey of 150 hospital decision-makers, it ranked as the second largest expense in 2015, with labour being the biggest cost. In 2020, however, it took the top spot, thanks to rising prices for overnight shipping, tighter operating margins, increased merger and acquisition activity, and an evolving regulatory landscape that rewards quality over volume. Today's healthcare supply chain challenges are far reaching and majorly involve medical technology and equipment. The Cardinal Health/SERMO report estimated a US\$ 5 billion loss across the global healthcare industry each year just because of high-value medical devices. Moreover, the integration of electronic health record (EHR) systems, including specialised solutions for fields like orthopaedics, makes the already complex IT systems even more unwieldy for many providers.

When prescription drugs, medical devices and other medical supplies fall into an optimised supply chain, healthcare providers see their costs

lowered, their revenues enhanced and, most importantly, their quality of care improved. To optimise the supply chain, providers need to address the biggest challenges that they face.

Overnight shipping: In the healthcare industry, there is no debate about the necessity of overnight shipping. Unexpected situations occur and hospitals have a responsibility to address them—no matter what the cost. One or two overnight orders will not have a detrimental impact on a provider's bottomline. However, when this occurs regularly—as it inevitably will—providers start seeing significant losses in their revenues. For example, FedEx will charge nearly US\$ 100 to send a package weighing 20kg within a 250km radius. This cost will only escalate for further distances or heavier items. With better inventory software, providers can completely eliminate their overnight shipping costs that are caused by stock outs. While some emergency items may need to be overnighted on an occasional basis, the regularity of the situation will drop dramatically.

Hidden costs: Another challenge that healthcare providers face is the hidden cost of every product. Most providers have historically just looked at the product cost and the shipping cost. But there are additional expenses, such as inventory holding. Providers need to plan their budget around total landed supply costs. They need to be aware of the losses they will incur from the unavoidable aspect of expired products and excess supplies. They also need to look at product standards and the variations in purchase prices. This is where quality healthcare software comes in, as it can help providers take a more holistic approach to pricing their products, taking into account the costs associated with moving and managing supplies.

Shortage of drugs: Because of the unpredictability that is part and parcel of the healthcare industry, drug